

The telegraph

In 1820, Hans Christian Ørsted discovered that electricity could create magnetism. From this, William Sturgeon and Joseph Henry developed the electromagnet—in which an electric current creates a strong magnetic field. In turn, this led Samuel Morse to resume his experiments in electricity and, eventually, a new way of communicating was born.

The position of the movable arm represents different letters and numbers.



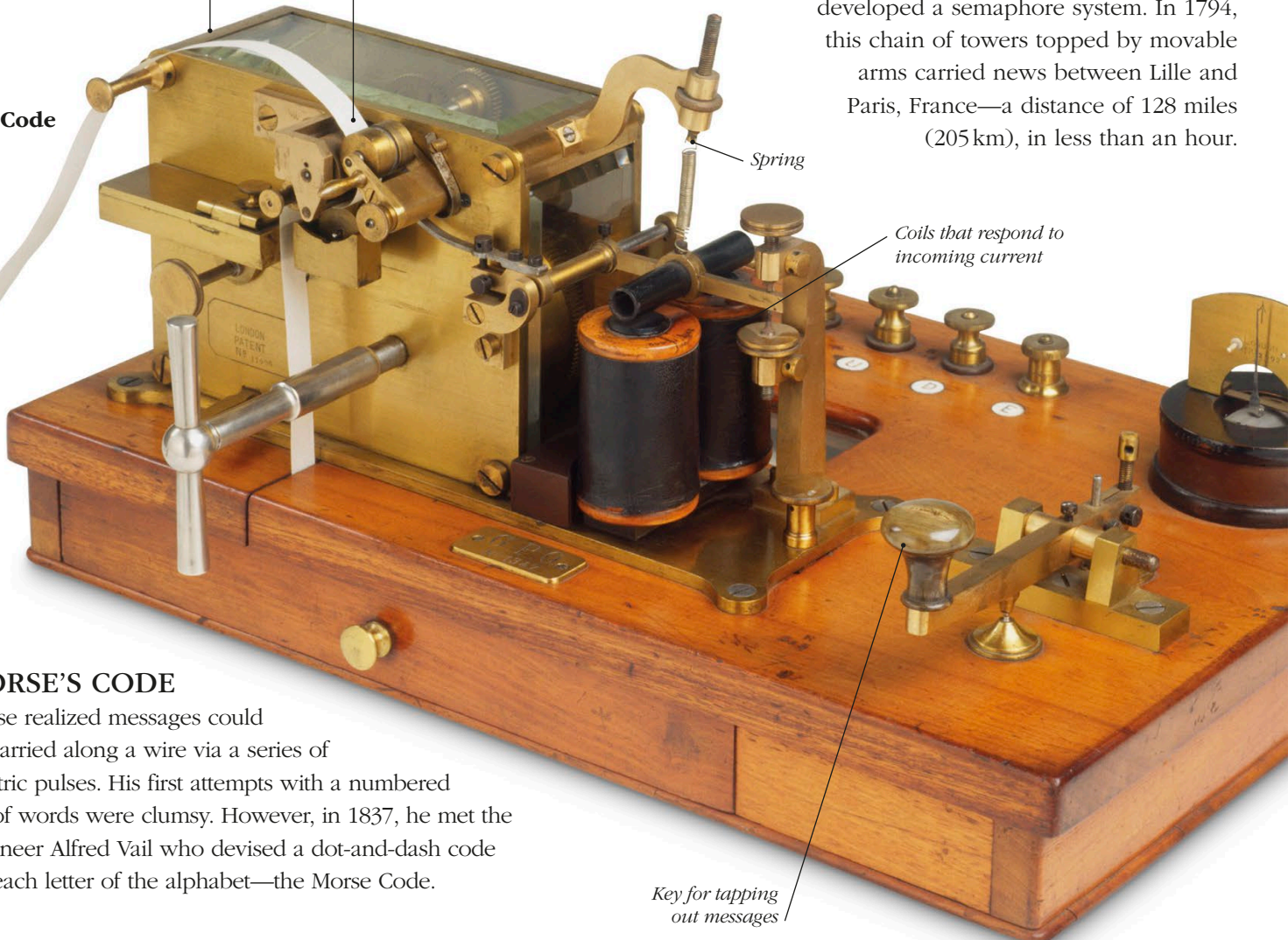
SYNCHRONIZED SEMAPHORE

The French inventor Claude Chappe developed a semaphore system. In 1794, this chain of towers topped by movable arms carried news between Lille and Paris, France—a distance of 128 miles (205 km), in less than an hour.

Stylus embosses dots and dashes on the paper tape.

Wheel drives tape through the machine.

Morse Code device



MORSE'S CODE

Morse realized messages could be carried along a wire via a series of electric pulses. His first attempts with a numbered list of words were clumsy. However, in 1837, he met the engineer Alfred Vail who devised a dot-and-dash code for each letter of the alphabet—the Morse Code.

Key for tapping out messages